# FUNDAMENTALS OF ELECTRICAL & ELECTRONICS ENGINEERING LAB

**Suggested Practicals/Exercises:**
The practical in this section are PrOs (i.e. sub-components of the COs) to be developed and assessedin the student for the attainment of the competency.

| S. No. | Practical Outcomes (PrOs) | Approx. Hrs. |
|---|---|---|
| 1. | Determine the permeability of magnetic material by plotting its B-H curve. | 02* |
| 2. | Measure voltage, current and power in 1-phase circuit with resistive load. | 02* |
| 3. | Measure voltage, current and power in R-L series circuit. | 02* |
| 4. | Determine the transformation ratio (K) of 1-phase transformer. | 02 |
| 5. | Connect single phase transformer and measure input and output quantities. | 02 |
| 6. | Make Star and Delta connection in induction motor starters and measure the line and phase values. | 02 |
| 7. | Identify various passive electronic components in the given circuit | 02 |
| 8. | Connect resistors in series and parallel combination on bread board and measure its value using digital multimeter. | 02 |
| 9. | Connect capacitors in series and parallel combination on bread board and measure its value using multimeter. | 02* |

| S. No. | Practical Outcomes (PrOs) | Approx. Hrs. |
|---|---|---|
| 10. | Identify various active electronic components in the given circuit. | 02 |
| 11. | Use multimeter to measure the value of given resistor. | 02 |
| 12. | Use LCR-Q tester to measure the value of given capacitor and inductor. | 02 |
| 13. | Determine the value of given resistor using digital multimeter to confirm with colour code. | 02* |
| 14. | Test the PN-junction diodes using digital multimeter. | 02* |
| 15. | Test the performance of PN-junction diode. | 02 |
| 16. | Test the performance of Zener diode. | 02 |
| 17. | Test the performance of LED. | 02 |
| 18. | Identify three terminals of a transistor using digital multimeter. | 02 |
| 19. | Test the performance of NPN transistor. | 02* |
| 20. | Determine the current gain of CE transistor configuration. | 02 |
| 21. | Test the performance of transistor switch circuit. | 02 |
| 22. | Test the performance of transistor amplifier circuit. | 02 |
| 23. | Test Op-Amp as amplifier and Integrator | 02 |
| | **Total** | **46** |

**References:**


1. Ritu Sahdev, Basic Electrical Engineering, Khanna Publishing House, 2018
2. Mittle and Mittal, Basic Electrical Engineering, McGraw Education, New Delhi, 2015, ISBN : 978-0-07-0088572-5
3. Saxena, S. B. Lal, Fundamentals of Electrical Engineering, Cambridge University Press, latest edition ISBN : 9781107464353
4. Theraja, B. L., Electrical Technology Vol – I, S. Chand publications, New Delhi, 2015, ISBN: 9788121924405
5. Theraja, B. L., Electrical Technology Vol – II, S. Chand publications, New Delhi, 2015, ISBN: 9788121924375
6. Jegathesan, V., Basic Electrical and Electronics Engineering, Wiley India, New Delhi, 2015, ISBN : 97881236529513
7. Sedha, R.S., A text book of Applied Electronics, S.Chand ,New Delhi, 2008, ISBN-13: 978-8121927833
8. Malvino, Albert Paul, David, Electronics Principles, McGraw Hill Eduction, New Delhi,2015,ISBN-13: 0070634244-978
9. Mehta, V.K., Mehta, Rohit, Principles of Electronics, S. Chand and Company, New Delhi, 2014,ISBN-13-9788121924504
10. Bell Devid, Fundamental of Electronic Devices and Circuits, Oxford University Press, New Delhi 2015 ISBN : 9780195425239


**Suggested Softwares/Learning Websites:**

a. en.wikipedia.org/wiki/Transformer
b. www.animations.physics.unsw.edu.au//jw/AC.html
c. www.alpharubicon.com/altenergy/understandingAC.htm
d. www.electronics-tutorials
e. learn.sparkfun.com/tutorials/transistors
f. www.pitt.edu/~qiw4/Academic/ME2082/Transistor%20Basics.pdf
g. www.technologystudent.com/elec1/transis1.htm
h. www.learningaboutelectronics.com
i. www.electrical4u.com

**Course Outcomes:**

At the end of the course student will be able to:

1. Understand basic principle and operation of electric circuits and machines.
2. Solve basic problems related to electrical circuits and machines. Explain the operation of different electrical technologies.
3. Demonstrate an understanding of the control systems.
4. Understand the basic circuit elements
5. Understand different types of signal waveforms.
6. Understand logic gates and apply them in various electronic circuits.
7. Understand the basic concepts of op-amps, and their applications.
8. Use relevant electric/electronic protective devices safely.

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**DIPLOMA WING**
**RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL**

*SEMESTER II – GROUP 'A'*

| | | |
|---|---|---|
| COURSE TITLE | : | FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING |
| PAPER CODE | : | 7355 |
| SUBJECT CODE | : | 204 |
| TREORY CREDITS | : | 03 |
| PRACTICAL CREDITS | : | 02 |

**Course Objectives:**

To provide basic knowledge of the different elements and concepts of electrical engineering field andto learn basic concepts of various active and passive electronic components, Signals, Op-Amp and their applications, Digital Electronics and their applications to help students deal with electrical and electronics engineering principles and applications in industrial processes of different fields.

**Course Content:**

**UNIT I** Overview of Electronic Components & Signals:

Passive Active Components: Resistances, Capacitors, Inductors, Diodes, Transistors, FET, MOS and CMOS and their Applications. Signals: DC/AC, voltage/current, periodic/non-periodic signals, aver- age, rms, peak values, different types of signal waveforms, Ideal/non-ideal voltage/current sources, independent/dependent voltage current sources.

**UNIT II** Overview of Analog Circuits:

Operational Amplifiers-Ideal Op-Amp, Practical op amp, Open loop and closed loop configurations, Application of Op-Amp as amplifier, adder, differentiator and integrator.

**UNIT III** Overview of Digital Electronics: Introduction to Boolean Algebra, Electronic Implementa- tion of Boolean Operations, Gates-Functional Block Approach, Storage elements-Flip Flops-A Func- tional block approach, Counters: Ripple, Up/down and decade, Introduction to digital IC Gates (of TTL Type).

**Unit IV** Electric and Magnetic Circuits:

EMF, Current, Potential Difference, Power and Energy; M.M.F, magnetic force, permeability, hyster- esis loop, reluctance, leakage factor and BH curve; Electromagnetic induction, Faraday's laws of electromagnetic induction, Lenz's law; Dynamically induced emf; Statically induced emf; Equations of self and mutual inductance; Analogy between electric and magnetic circuits.

**Unit V** A.C. Circuits:

Cycle, Frequency, Periodic time, Amplitude, Angular velocity, RMS value, Average value, Form FactorPeak Factor, impedance, phase angle, and power factor; Mathematical and phasor representation ofalternating emf and current; Voltage and Current relationship in Star and Delta connections; A.C in resistors, inductors and capacitors; A.C in R-L series, R-C series, R-L-C series and parallel circuits; Power in A. C. Circuits, power triangle.

**Unit VI** Transformer and Machines: General construction and principle of different type of trans- formers; Emf equation and transformation ratio of transformers; Auto transformers; Construction and Working principle of motors; Basic equations and characteristic of motors.

**References:**


1. Ritu Sahdev, Basic Electrical Engineering, Khanna Publishing House
2. Mittle and Mittal, Basic Electrical Engineering, McGraw Education, New Delhi, 2015, ISBN : 978-0-07-0088572-5
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